

# Unit 5 Pre-Test

## The Pythagorean Theorem

Standards assessed: NC.8.G.6, NC.8.G.7, NC.8.G.8

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Period: \_\_\_\_\_

### SCORE 3.0 Basic Understanding

**NC.8.G.7** Apply the Pythagorean theorem to find the hypotenuse.

• I can use  $a^2 + b^2 = c^2$  to find the hypotenuse of a right triangle.

Match each vocabulary term to its definition, then find each hypotenuse.

|                           |                                                                  |
|---------------------------|------------------------------------------------------------------|
| C ▾ hypotenuse ✓          | A. The two sides that form the right angle.                      |
| A ▾ legs ✓                | B. $a^2 + b^2 = c^2$ , relating the sides of a right triangle.   |
| B ▾ Pythagorean theorem ✓ | C. The side opposite the right angle; the longest side.          |
| D ▾ right triangle ✓      | D. A triangle that contains one $90^\circ$ angle.                |
| E ▾ square root ✓         | E. A number that, multiplied by itself, gives a given value.     |
| F ▾ converse ✓            | F. If $a^2 + b^2 = c^2$ , then the triangle is a right triangle. |

#### Pythagorean theorem — reference

$$a^2 + b^2 = c^2$$

$c$  is the hypotenuse (opposite the right angle);  $a$  and  $b$  are the legs.

1. Find  $c$

Legs 3 and 4. Find the hypotenuse.

$c =$

answer: 5

2. Find  $c$

Legs 6 and 8. Find the hypotenuse.

$c =$

answer: 10

3. Find  $c$

Legs 5 and 12. Find the hypotenuse.

$c =$

answer: 13

4. Find  $c$

Legs 8 and 15. Find the hypotenuse.

$c =$

answer: 17

5. Find  $c$

Legs 9 and 12. Find the hypotenuse.

$c =$

answer: 15

6. Find  $c$

Legs 7 and 24. Find the hypotenuse.

$c =$

answer: 25

## SCORE 4.0 Target Content

**NC.8.G.7** Find a missing leg and estimate non-perfect square roots.

• *I can find a missing leg and round answers to the nearest tenth when needed.*

**Find each missing side. Round to the nearest tenth where asked.**

7. Find a leg

One leg 6, hypotenuse 10. Find the other leg.

$a =$

answer: 8

8. Find a leg

One leg 5, hypotenuse 13. Find the other leg.

$a =$

answer: 12

9. Find a leg

One leg 9, hypotenuse 15. Find the other leg.

$a =$

answer: 12

10. Nearest tenth

Legs 2 and 3. Find the hypotenuse to the nearest tenth.

$c \approx$

answer: 3.6

11. Nearest tenth

12. Nearest tenth

Legs 5 and 7. Find the hypotenuse to the nearest tenth.

$c \approx$

answer: 8.6

One leg 4, hypotenuse 9. Find the other leg to the nearest tenth.

$a \approx$

answer: 8.1

SCORE **5.0** Advanced Content

**NC.8.G.8** Apply the theorem to the coordinate plane and real-world problems.

• *I can find the distance between two points and solve real-world right-triangle problems.*

**Solve each problem.**

**13.** Distance

Find the distance between (0, 0) and (6, 8).

$d =$

answer: 10

**14.** Distance

Find the distance between (1, 2) and (4, 6).

$d =$

answer: 5

**15.** Apply

A 13 ft ladder reaches 12 ft up a wall. How far is the base from the wall?

$=$

answer: 5 ft

**16.** Apply

A rectangle is 9 by 12. Find the length of its diagonal.

$d =$

answer: 15

**17.** Distance

Find the distance between (-2, -1) and (3, 11).

$d =$

answer: 13

**18.** Apply

A park is a right triangle with legs 20 m and 21 m. Find the longest side.

$=$

answer: 29 m